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Acoustic Black Hole

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ABSTRACT

The term “black hole” is associated with the cosmological phenomenon by the gravitational field of a very massive object. The force of attraction is so great that even light can't get out of this trap. The General Theory of Relativity shows that this capture is associated with slowing down the speed of light, or, more generally, with slowing down the flow of time, from the viewpoint of an external observer. In classical wave physics, there are well known examples of slow propagation of waves. The speed of sound in gas decreases as the temperature decreases in proportion to the square root of the temperature. The speed of bending waves in the rod is proportional to the square root of its thickness. The speed of shear waves in ocean sedimentary rocks depends on how much precipitation has been deposited. The question that, as a result, gave rise to the branch of Black Holes in acoustics is as follows. Is it possible to slow down the propagation of a wave on a way of finite length, so that the propagation time would be infinite? However, the change must not be too sharp, so that there would be no reflections from the intermediate sections. The answer is - Yes. Based on aforementioned discussion, this lecture start from the birth and growth of acoustic black hole (ABH) research with a historical review. Then, a couple of studies done by the authors are explained in theoretical, numerical and experimental aspects. This lecture ends with the efforts and trials for practical application of ABH to real-world problems with a short concluding remark on perspectives and opportunities of ABH for the researchers in structural vibrations, vibro-acoustics, and wave physics.

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1. INTRODUCTION

ABH research has now become quite popular. The attractiveness of this direction is related, on the one hand, to the potential efficiency of vibration absorption by withdrawing vibration energy and concentrating it in a small area where it can be absorbed by a small amount of absorbing material. On the other hand, the mechanism of slowing down the waves to such a speed of propagation that the propagation time of the waves increases significantly. The possibility of a smooth decrease in the propagation speed of a bending wave in a smoothly pointed rod without reflection from intermediate sections was shown in [1] in 1988. After that, there was a long period of stagnation, most likely due to the lack of information about this work. The acceleration of research, expressed in an increase in publications in journals and conference proceedings, began about 10 years ago. In a recent review [2] a detailed picture of the history of research, current achievements, and even a thorough classification of various trends in the theory, experiment, and applications of acoustic black holes is given.

This report focuses mainly on the ideas of the authors of the report who are actively researching ABH and partially supplements the review article [2]. Section 2 of the report describes the physical idea of ABH and describes the first laboratory experiments on vibration absorption using ABH. The Rayleigh problem [3] on a string with a variable linear mass is mentioned, in which the distribution of the velocity of propagation of bending waves leads to the BH effect. An alternative possibility of creating a deceleration of the propagation speed in the rod by means of transverse slots in it is noted. In Section 3, the history of ABH is briefly reviewed, and a couple of recent works done by the authors are summarized. A concept of spiral ABH whose baseline is spiral shape is described, and numerical and experimental investigations on vibration damping and noise reduction using the spiral ABH are explained. In addition to numerical and experimental studies, two mathematical theories of ABHs are briefly introduced, aiming at understanding better the physics of wave motions in ABH and designing the ABH in a fast and accurate way. Finally, a short concluding remark is given in Section 4.

2. BIRTH AND IDEA OF ABH

The starting point in the investigation was not to develop an effective broadband vibration and sound absorber. The question, that gave start of the development of ABH, is as follows. Is it possible to slow down the propagation velocity of a wave of some nature on a segment of finite length H (see Figure 1) so, that the propagation time on this segment would be infinite? However, the velocity variation should not be too sharp, so that there are no reflections from the intermediate sections.

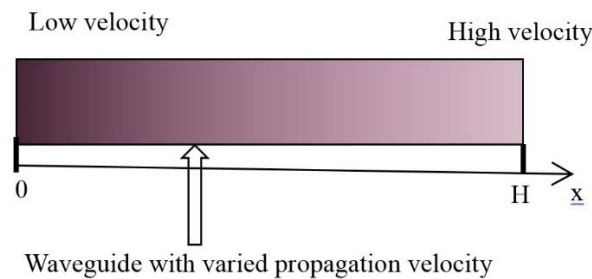


Figure 1: Segment of waveguide with varied propagation velocity

The propagation of a harmonic wave of any nature (sound, electromagnetic waves, waves on the surface of water, etc.) in a homogeneous medium is described by the Equation 1:

$$P(x, t) = A \exp(-i\omega t + ikx) \quad (1)$$

ω - frequency, $k = \omega/c$ - wave number, c - wave phase velocity, x - direction of propagation. If the propagation velocity depends on the coordinate: $c = c(x)$, the previous formula should be rewritten in such a way:

$$P(x, t) = A \exp \left(-i\omega t + i\omega \int \frac{dx}{c(x)} \right) \quad (2)$$

The propagation time T from x_0 to x_1 is equal to this integral:

$$T(x_0, x_1) = \int_{x_0}^{x_1} \frac{dx}{c(x)} \quad (3)$$

Let's suppose the propagation velocity tends to zero at x_1 tending to zero. For certainty, take the power dependence:

$$c(x) = c_0 \left(\frac{x}{x_0} \right)^n \quad (4)$$

Substituting equation (4) into equation (3) we get simple dependencies of the propagation time on x_1 and n . Below are graphical dependencies the propagation velocities (left) and propagation time for $x_1 = 0$.

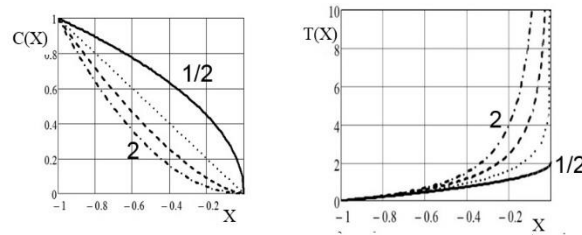


Figure 2: Propagation velocity (left) and propagation time (right) dependencies. n is equal to $1/2$, 1 , $3/2$, 2

At $n < 1$, the time to reach the point $x = 0$ turned out to be finite. At $n \geq 1$, the wave propagates infinitely up to $x = 0$. It will never reach the end and, as a consequence, will not be reflected. We obtain an ideal wave absorber – BLACK HOLE - in which the absorption is does not related to the material absorption. Specific implementations of the black hole principle have certain limitations on the frequency range and manufacturing accuracy.

And what about the condition for smooth deceleration of the propagation velocity? The answer can be obtained simply on the basis of quasi-classical (WKB) approximation: the wavelength λ should change quite slowly: $\partial\lambda(x)/\partial x < 1$ or $\partial c(x)/\partial x < \omega$ – the rate of change of propagation velocity should be less than the frequency.

The simplest example, in sense of its manufacturing, is a sharpened rod.

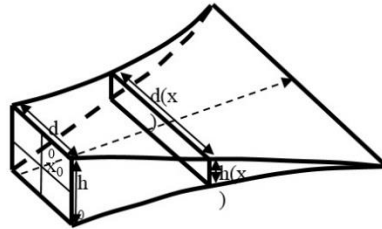


Figure 3: Simplest black hole – parabolically sharpened rod

The propagation velocity of bending wave in the rod is inversely proportional to the square root of its thickness $c \sim \sqrt{h}$. This means that if the thickness of the rod is reduced according to the quadratic law $h \sim x^2$, or faster, the propagation velocity will decrease quite quickly. In the article [1] it was noted that such a rod would be an ideal vibration absorber. A wave that has entered the initial, thick section will propagate infinitely in time to the sharpened edge. Therefore, it will never

reach it and will not be reflected. The reflection coefficient from a properly sharpened rod turns out to be zero. So, it turned out the Vibration Black Hole.

Of course, this ideal was practically unattainable. We can't bring the thickness strictly to zero. But despite this, the design is useful. As the wave moves towards the sharpened section, the energy flow is preserved. The energy density increases. A small addition of absorbing material creates a very effective absorption.

The parabolically sharpened rod was manufactured and some experiment were conducted. The impact excited two rods – uniform and parabolically sharpened. A small piece of plasticine, or a piece of paper, was attached to the end of the edge. The impact sound was registered and the result was spectacular. The corresponding time dependencies are shown in Fig. 4. The rod with BH sounds about 10 ms. A uniform rod sounds infinitely long.

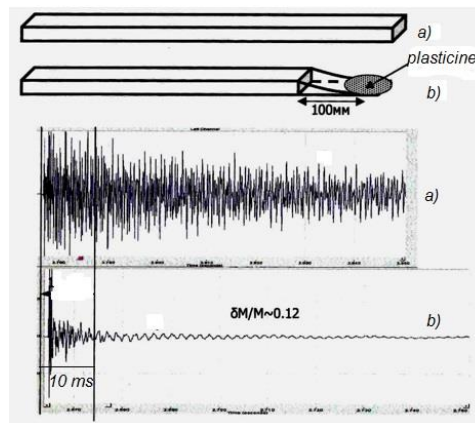


Figure 4: Sounds, excited by impact: a) - uniform rod, b) – parabolically sharpened rod

Next, several BH were produced – in the form of disks with parabolically sharpened edges [4]. One of the experimental results are shown below (Figure 5). A thick steel plate 9.5 mm thick was excited by the impact. Vibrations were registered. Then a black hole in the form of a disk with a small layer of absorbing material (plasticine) was attached to the plate. The vibrations were recorded again. The image below shows the vibration spectra without BH (blue) and with BH (red) in the frequency band up to 5 kHz. Most of the resonant peaks are significantly suppressed.

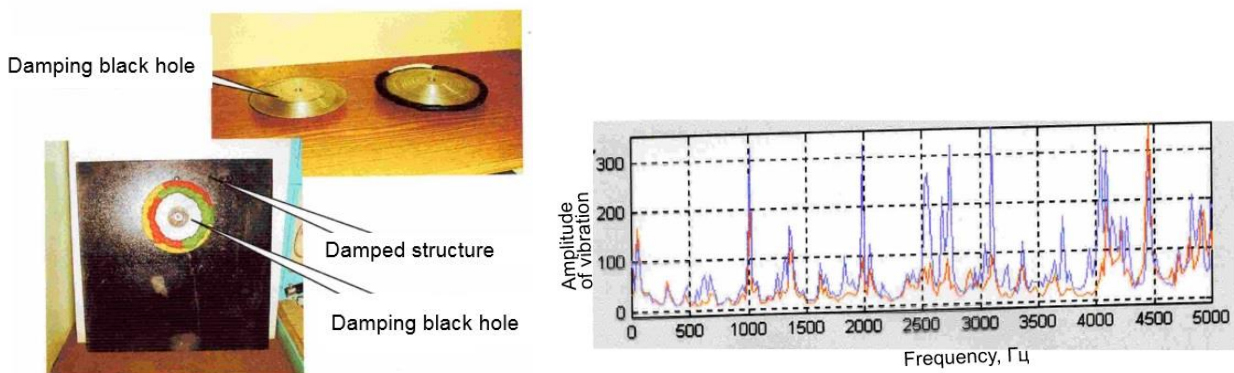


Figure 5: Thick plate with damping black hole (left) and plate vibrational spectra without (blue) and with black hole (red) (right)

Then several designs resembling BH will be marked in the talk [5]. Some pictures taken from the patents are given in Fig. 6.

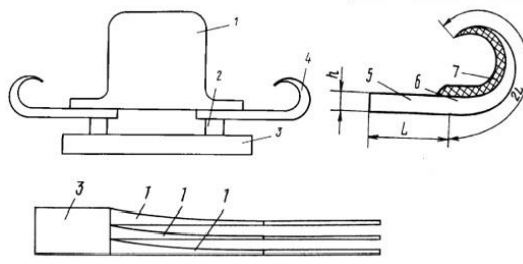


Figure 6: Constructions of vibration absorbers

A medium with a linear law changing propagation speed was considered by the outstanding founder of modern acoustics, **Lord Strutt (Baron Rayleigh)** (see Figure 7). Rayleigh considered the transverse vibrations of a string whose density changes inversely with the square of the coordinate. In this case, the string tension, of course, does not depend on the coordinate [3]. With this density distribution, the velocity of propagation of transverse waves on the string is proportional to the coordinate.



$$\frac{\partial^2 y}{\partial x^2} + n^2 x^{-2} y = 0,$$

$$y = A x^{1/2 + im} + B x^{1/2 - im},$$

$$m^2 = n^2 - \frac{1}{4}.$$

Figure 7: Rayleigh, equation and solutions of transverse displacement of string

He wrote down an equation for the transverse displacement of the string y with variable density and got a set of solutions in the form of two **power functions**. This formula contains all the signs of BH. Of course, the tendency of the linear mass of the string to infinity at $x > 0$ is physically unrealizable. Therefore, Rayleigh did not pay attention to the fact that he came up with the BH.

Using a bending oscillating rod as the initial work piece for the production of BH gives us more opportunities than a string. To slow down of propagation velocity of the bending wave, it is not necessary to increase the linear mass, as Rayleigh did in the case of a string. Moreover, in a sharpened rod, the linear mass is decreased in proportion to the thickness. The reduction of the propagation velocity is due to the fact that the bending stiffness is decreased even faster – in proportion to the cube of thickness. And what happens if the bending stiffness of the rod is reduced, but the linear mass is not reduced? It is clear that propagation velocity of the bending wave will be even less. The device of such a modified BH may look like this (see Figure 8). This is a homogeneous rod with transverse slits. Slits reduce bending stiffness and practically do not change the linear mass. The BH threshold frequency is reduced. The requirements for manufacturing accuracy is reduced [6].

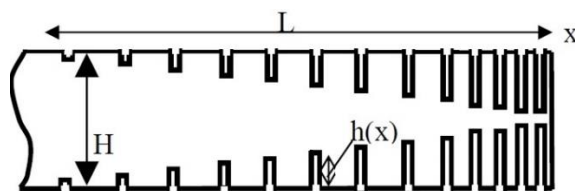


Figure 8: Modified BH – rod with transversal slits

3. GROWTH OF ABH AND RECENT ADVANCES

In this section, we briefly review the history of ABH and introduce a couple of recent works done by the authors. As explained in the previous section, ABH was born in 1988. However, there was more than ten years of dormant period of ABH until it was revisited by Victor Krylov in 21st century. Then, for recent ten years, we've been going through the blooming period.

As illustrated in Fig. 9, the number of journal papers and conference talks on ABH is rapidly increasing for recent ten years, and the studies related to ABH are covering a wide range of interesting topics such as the physical principles of ABH, the applications to energy harvesting and bio-inspired systems, and the elastic wave manipulation through ABH-based metamaterials, as well summarized in the aforementioned review paper [2]. Here, we provide the list of most recent studies by adding 14 articles [10-23] published after the review paper.

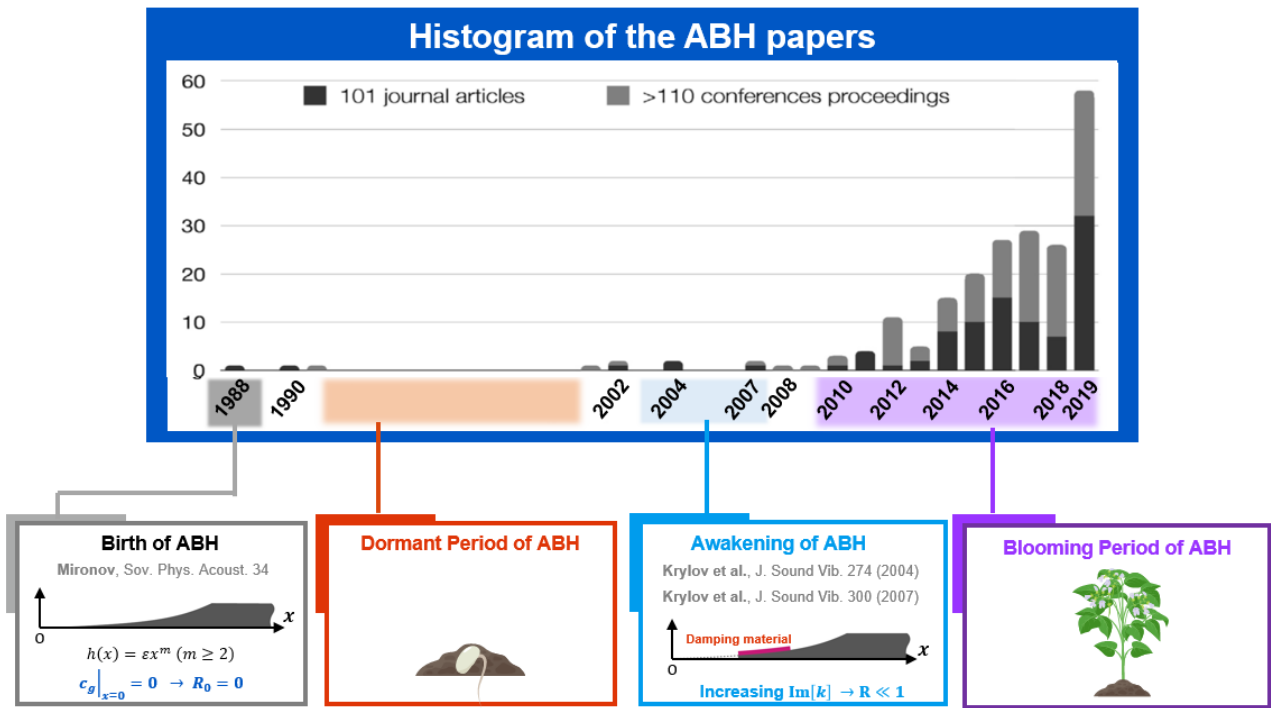


Figure 9: Histogram of ABH papers (top) and brief history of ABH research divided into four stages (bottom)

Among more than 100 interesting and important works during recent ten years, in this section, we briefly summarize a couple of works on ABH conducted by the authors in theoretical, numerical, and experimental aspects.

Firstly, numerical and experimental investigations on vibration damping and noise reduction using the spiral ABH are explained [7, 8]. It is well known that the damping performance increases as the length of an ABH increases. However, the increased length also increases the space occupied by ABH. The spiral ABH concept was proposed from the following question in the viewpoint of practical application. How can we enhance the damping performance and save the space occupied by ABH, simultaneously? To modify the baseline shape of the ABH from a straight line into a curve can be a simple solution to the question. Among various curves, an Archimedean spiral curve was utilized in the ABH baseline as shown in Fig. 10 as it is able to minimize the possibility of physical contact between baselines during dynamic deformation.

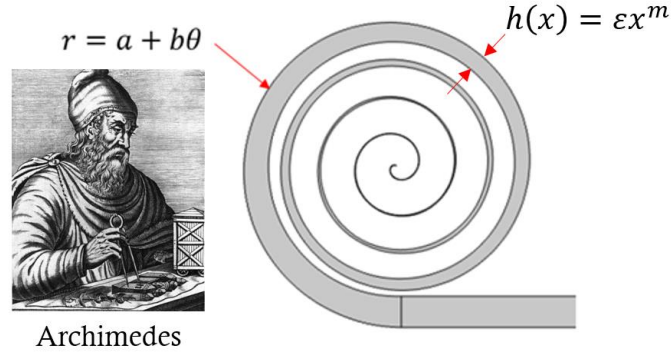


Figure 10: Schematic of Archimedean spiral ABH

To check if the damping performance of the spiral ABH is similar to that of straight ABH, the driving point mobilities of straight and spiral ABHs were investigated numerically and those are compared in Fig. 11. As shown in the results, almost same peak levels were obtained, which implies that the standard (straight) ABH can be replaced by the spiral one without sacrifice in the performance.

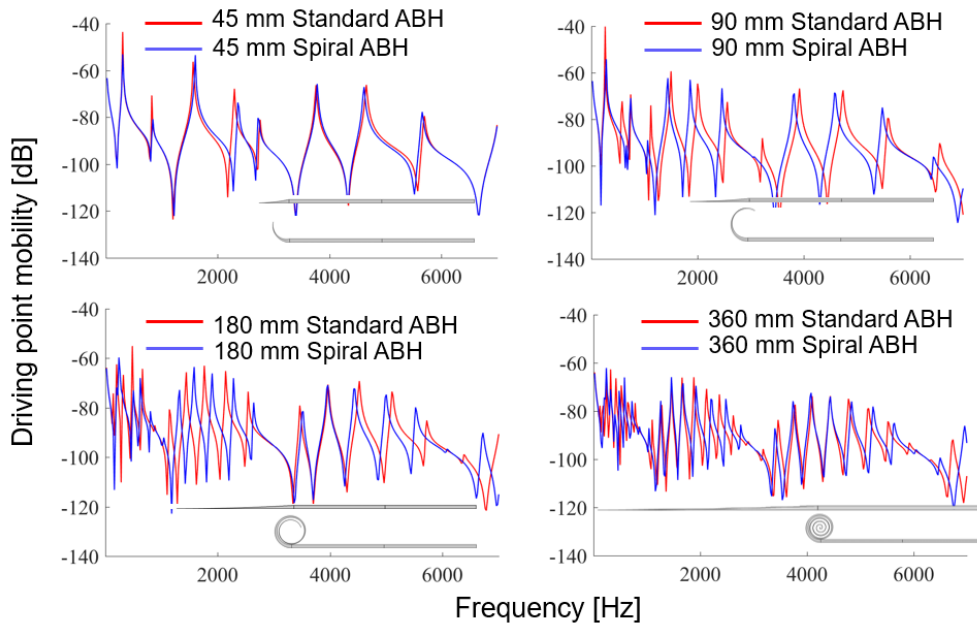


Figure 11: Driving point mobilities of straight ABHs and spiral ABHs

In order to realize the concept of spiral ABH, it was necessary to fabricate the spiral ABH and validate its damping performance. However, it was a challenging task because the tip was very thin, the thickness should follow a certain power-law profile, and the baseline must be curled by an Archimedean spiral. After various trials for manufacturing, the spiral ABH could be precisely fabricated using a wire electrical discharge machine. Its damping performance was investigated experimentally by the impact hammer test. In the results shown in Fig. 12, large peak reductions more than 20 dB could be achieved using the spiral ABH with damping material treatment near the tip region.

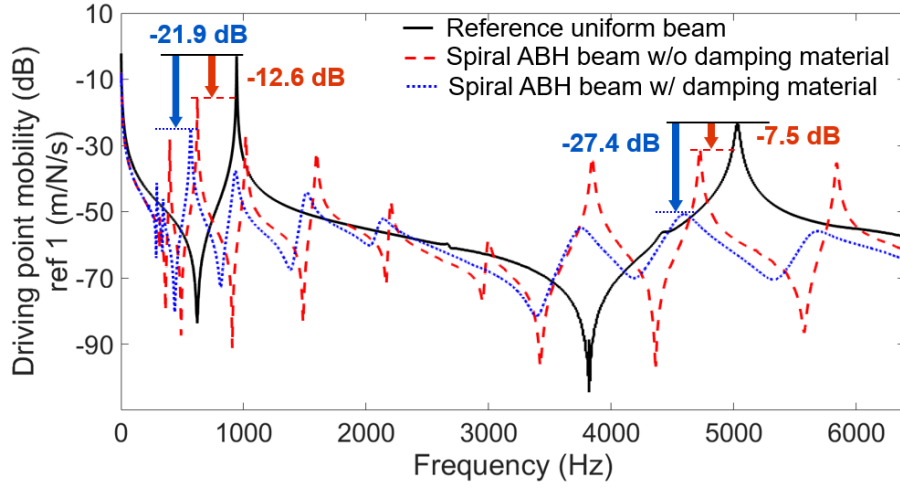


Figure 12: Driving point mobility of reference beam, 360mm spiral ABH beams with and without damping material

In addition to numerical and experimental studies, two mathematical theories of ABHs were developed. One is about the derivation of exact solution for straight ABHs with arbitrary power ‘m’ [9], and the other is about semi-analytical method for curved ABHs. Using the theories, reflection coefficient and cut-on frequency for any shapes of straight ABH or spiral ABHs can be calculated, leading to better understand the physics of wave motions in ABH and to design the ABHs in a fast and accurate way.

The mathematical theory for straight ABHs is first summarized briefly. Euler-Bernoulli equation for acoustic black holes can be described by Equation 5.

$$x^{2m} \frac{d^4 w}{dx^4} + 6mx^{2m-1} \frac{d^3 w}{dx^3} + 3m(3m-1)x^{2m-2} \frac{d^2 w}{dx^2} - \xi w(x) = 0 \quad (5)$$

where m is the power of ABH profile and ξ is $\frac{12\rho\omega^2}{E\varepsilon^2}$. The Euler-Bernoulli equation for the ABH can be reformulated into the form of a generalized hypergeometric differential equation, shown in the following equation.

$$\left\{ \vartheta \left(\vartheta + \frac{3-2m}{4-2m} - 1 \right) \left(\vartheta + \frac{2+m}{4-2m} - 1 \right) \left(\vartheta + \frac{1+m}{4-2m} - 1 \right) - z \right\} w(z) = 0 \quad (6)$$

where $\vartheta = z \frac{d}{dz}$ and $z = \frac{\xi}{(4-2m)^4} x^{4-2m}$. Then, the exact solution for the equation is derived in terms of generalized hypergeometric functions (pFq) where $p=0$ and $q=3$ and the functions (pFq) is shown in Equation.7.

$${}_pF_q(a_1, a_2, \dots, a_p; b_1, b_2, \dots, b_q; z) = \sum_{n=0}^{\infty} \frac{(a_1)_n (a_2)_n \dots (a_p)_n}{(b_1)_n (b_2)_n \dots (b_q)_n} \frac{z^n}{n!} \quad (7)$$

where $(a)_n = a(a+1) \dots (a+n-1)$ is the Pochhammer symbol.

The exact solution can be used to calculate reflection coefficient of a ABH-attached structure. Consider an example illustrated in Fig. 13(a): an ABH is attached to a uniform beam. Free boundary conditions were given at both ends ($x = x_1$ and $x = x_3$) and a harmonic excitation with a unit amplitude was applied to the beam at $x = x_f$. In this example, the displacement fields of the ABH-attached structure using the exact solution and FEM were compared in Fig. 13(b). In addition, the

author could examine the reflection coefficients obtained from the approximate method based on the geometrical acoustics and the present exact theory, as compared in Fig. 13(c). The results in Figs. 13 (b) and (c) show that by using the exact solution for any shapes of ABH, it is possible to calculate the displacement field and reflection coefficient of ABH-attached structures in a fast and accurate way instead of using the time-consuming finite element simulations.

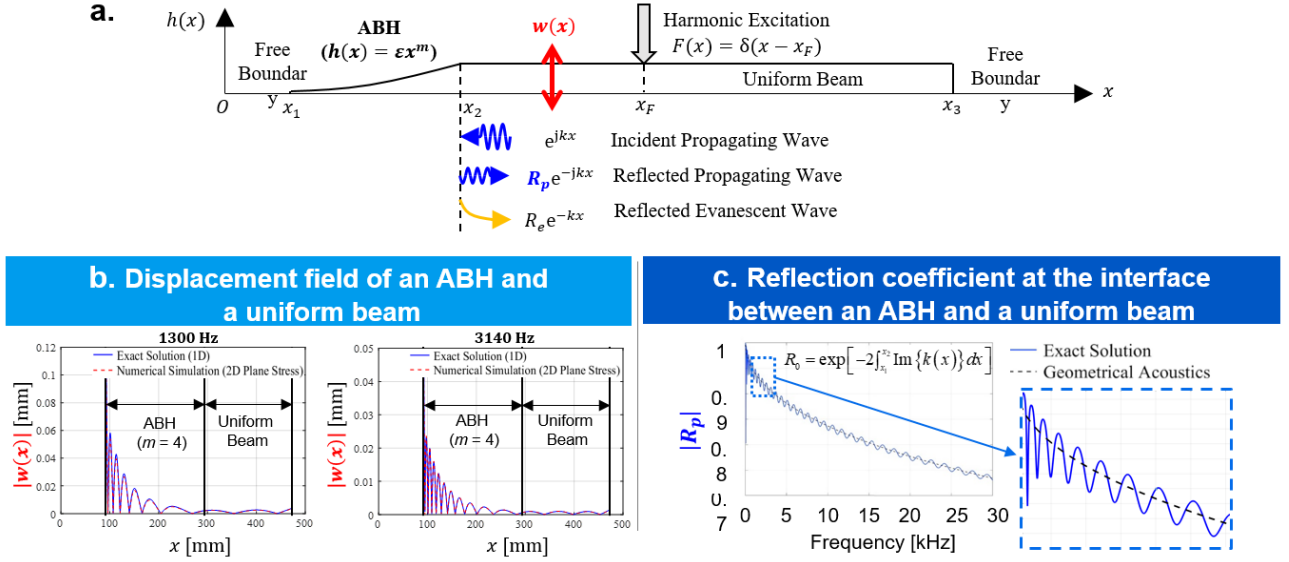


Figure 13. (a) A schematic of a uniform beam with an ABH. (b) Displacement fields and (c) reflection coefficient of the ABH-attached structure, obtained by the exact solution

In addition to the rigorous mathematical theory for straight ABHs, the first author applied the Love's theory of vibration of thin shells to the curved ABHs, which have varying thickness and arbitrary curvature, and the governing equations of wave motion in the curved ABHs could be derived as shown in Equations 8 and 9.

$$\frac{d}{ds} \left\{ EA \left(\frac{d\hat{u}}{ds} - \frac{\hat{w}}{R} \right) \right\} + EI \frac{dR}{ds} \left(\frac{d^2 \hat{w}}{ds^2} - \frac{1}{R^2} \frac{dR}{ds} \hat{u} + \frac{\hat{w}}{R} \right) + \rho A \omega^2 \hat{u} = 0 \quad (8)$$

$$-\frac{d^2}{ds^2} \left\{ EI \left(\frac{d^2 \hat{w}}{ds^2} - \frac{1}{R^2} \frac{dR}{ds} \hat{u} + \frac{\hat{w}}{R^2} \right) \right\} + \frac{EA}{R} \left(\frac{d\hat{u}}{ds} - \frac{\hat{w}}{R} \right) - EI \left(\frac{d^2 \hat{w}}{ds^2} - \frac{1}{R^2} \frac{dR}{ds} \hat{u} + \frac{\hat{w}}{R^2} \right) + \rho A \omega^2 \hat{w} = 0 \quad (9)$$

where $\hat{w}$ and $\hat{u}$ indicate normal and tangential deflection, respectively. But a simple analytic solution is not available since the flexural waves are coupled with the longitudinal waves. So, the governing equation was reformulated into an impedance equation that allows computers to calculate the reflection coefficients of curved ABHs, and Fig. 14 shows their reflection coefficients obtained by using the impedance equation for different curvatures and gap distances.

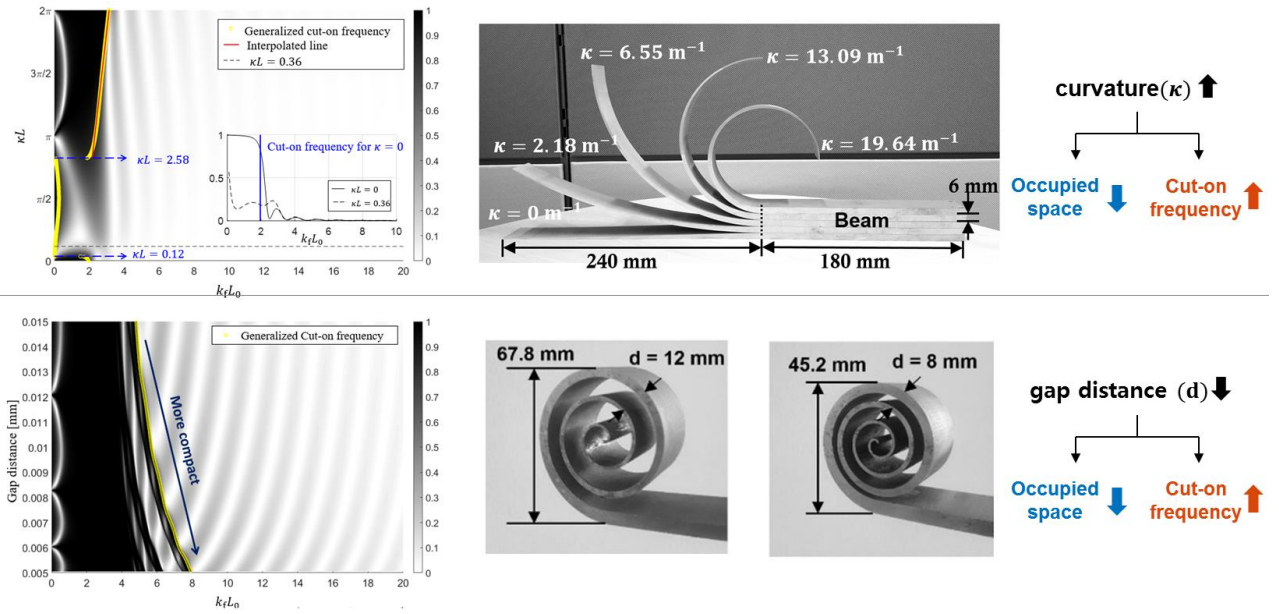


Figure 14. Reflection coefficients of Arc and Spiral ABHs depending on Curvature (κ) and Gap distance (d), obtained by the proposed theory for curved ABH

For a practical application of the curved ABH theory, the author investigated the effects of curvature and gap distance on the cut-on frequencies of arc and spiral ABHs. In Fig. 14, the yellow lines indicate the cut-on frequencies of arc and spiral ABHs. As shown in Fig. 14, increasing the curvature or decreasing the gap distance make the spiral ABH more compact, but at the same time, the cut-on frequencies also increased. So, it is important to control the size or curvature of ABH so as not to increase the cut-on frequency beyond the large peak frequencies in mechanical systems. By using the proposed mathematical theory for curved ABHs, it is possible to predict the cut-on frequency of a designed ABH before manufacturing it. In other words, the mathematical theory of curved ABHs can be used as a fast and accurate tool for design of a spiral ABH. Last not but least, a couple of examples regarding the application of ABHs will be given in the presentation.

4. CONCLUSIONS

In this paper, the authors reviewed history of ABH from the birth and idea to the growth and recent advances in theoretical, numerical, and experimental results. Among many works published for recent ten years, only a couple of the authors' works were briefly summarized with a reference of review paper [2]. There are increasing efforts and trials to apply the acoustic black holes to various industrial problems in home appliances, electrical power systems, rotating machineries, transportation systems and vehicles. As one of examples, the ABHs are developed to realize anechoic shock testing device that utilize in a defence sector [10]. Acoustic Black holes are still going through the period of growth and will reach the next stage of maturity in the near future. The authors hope that many researchers in the fields of wave physics, structural vibrations, and vibro-acoustics will be interested in ABH, leading to successful practical application to real-world problems.

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